

Homework 3

In this problem set, the following notations are used

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Problem 1. Let ρ be the density matrix of a qubit and let $U_{a,b} = \sigma_z^a \sigma_x^b$. The one-time pad $\text{OTP}_{a,b}(\rho)$ with keys $a, b \in \{0, 1\}$ is defined to be

$$\text{OTP}_{a,b}(\rho) = U_{a,b} \rho U_{a,b}^\dagger.$$

(a) Prove that for any qubit state ρ ,

$$\mathbb{E}_{a,b} \text{OTP}_{a,b}(\rho) = \frac{I}{2}.$$

- (b) It is possible to do interesting computation on the state ρ protected by the one-time pad without decoding. Show that if we apply an X gate on $\text{OTP}_{a,b}(\rho)$, we get $\text{OTP}_{a,b}(X\rho X)$.
- (c) Show that if we apply H on $\text{OTP}_{a,b}(\rho)$ we obtain the one-time pad state $\text{OTP}_{b,a}(H\rho H)$. Notice that the value of a, b are swapped in this case!

Solution. (a) Write the solution here.

- (b)
(c)

Problem 2. Assume that state is either $|\psi_0\rangle$ or $|\psi_1\rangle$ with equal probability and $|\langle\psi_0|\psi_1\rangle| = s > 0$. Design a positive-operator valued measure (POVM) $\{M_0, M_1, M_*\}$ with three outcomes 0, 1, and $*$ to distinguish the state such that

1. if the outcome is 0, the state must be $|\psi_0\rangle$,
2. if the outcome is 1, the state must be $|\psi_1\rangle$,
3. the probability of having $*$ as the outcome is the smallest possible. Note that in this case, the measurement does not tell us whether the state is $|\psi_0\rangle$ or $|\psi_1\rangle$, and it's allowed to answer "I don't know". So the goal here is to minimize the probability of outputting an inconclusive answer.

Solution.

Problem 3. Let ρ be a two-qubit state. Prove that the process of (1) applying a CNOT gate using the first qubit as the control and the second qubit as the target and (2) measuring the second qubit in Z basis is the same as measuring the state ρ directly using observable $Z \otimes Z$.

Solution.

Problem 4. In this problem, we consider a game between a quantum computer and a classical referee. Let $f : \{0, 1\}^n \rightarrow \{0, 1\}^{n-1}$ be a 2-to-1 function, meaning that every $y \in \{0, 1\}^{n-1}$ has exactly two distinct preimages. Suppose there is an efficient (with a polynomial number of gates) quantum circuit to compute f , but no efficient circuit that can produce from given $x \in \{0, 1\}^n$ an $x' \neq x$ such that $f(x') = f(x)$. Show that, like in the Simon's algorithm, the quantum computer can efficiently generate a uniformly random y and an associated n -qubit state $|\phi_y\rangle$. The quantum computer sends the string y to the classical referee and keeps $|\phi_y\rangle$. Next the referee flips a coin and asks the quantum computer to perform one of the following two tasks.

1. (Task 0). When asked, efficiently generate an x such that $f(x) = y$; and
2. (Task 1). When asked, efficiently sample uniformly from the set $\{a \in \{0, 1\}^n \mid a \cdot (x \oplus x') = 0 \pmod{2}\}$, where x, x' are the two preimages of y under f .

Show that the quantum computer can always succeed.

(Note: This game is used in a proof of quantumness protocol as classical computers cannot always succeed.)

Solution.

Problem 5. Let U be an n -qubit unitary operator. Consider the input state

$$\rho_{\text{in}} = |0\rangle\langle 0| \otimes \frac{I}{2^n}.$$

Apply the following operations to the state:

1. Apply a Hadamard gate H to the first qubit.

2. Apply a controlled- U gate, where the first qubit is the control qubit.
3. Apply another Hadamard gate H to the first qubit.
4. Measure the first qubit in the computational basis.

Define success as obtaining the measurement outcome 0 on the first qubit. Show that the success probability is

$$P(0) = \frac{1}{2} \left(1 + \operatorname{Re} \left[\frac{\operatorname{Tr}(U)}{2^n} \right] \right).$$

Solution.